

Finance Prompt 1: Pick one industry that has been disrupted by fintech (payments, lending, insurance, or another). Explain what the traditional model looked like, what replaced it, and who lost out in the change.

How Fintech Broke the Remittance Industry

Sending money home used to cost a fortune. For decades, migrant workers in the Gulf, including millions in the UAE, walked into a Western Union or MoneyGram branch, handed over cash, paid a fee of 7 to 10 percent, and waited days for the recipient to collect it. The global remittance market moves over USD 650 billion annually, and the fees alone were worth tens of billions. This essay examines how fintech companies dismantled that model, who benefited, and who lost.

The legacy remittance system operated through physical branch networks. Western Union at its peak operated over 500,000 agent locations worldwide. The fee structure had two visible components and one hidden one. The visible components were the flat transaction fee (often USD 5 to 15) and a percentage fee on the amount sent. The hidden component was the exchange rate markup: the company would offer a rate less favourable than the market rate, pocketing the difference. On a USD 200 transfer, the total cost, including the spread, could reach USD 14 to 20, or 7 to 10 percent.

The reason fees stayed high for decades is structural. Building and maintaining hundreds of thousands of physical locations in dozens of countries requires enormous capital. Regulatory compliance in each jurisdiction, anti money laundering obligations, and the logistics of handling physical cash all add cost. These barriers to entry protected incumbents: a new competitor would need to replicate the global branch network before it could compete on price.

Fintech companies attacked the cost structure by eliminating the branch network. Wise (formerly TransferWise), founded in 2011, routes transfers through a peer to peer matching system: instead of physically moving money across borders, it matches senders and recipients within each country, so the money effectively stays put. A user in the UAE sending dirhams to a family in India is matched with someone in India sending rupees the other direction. The actual cross border movement of funds is netted and minimised, which dramatically reduces the cost. Wise charges fees of roughly 0.5 to 1.5 percent, uses the mid market exchange rate (the real rate with no markup), and completes most transfers within hours rather than days. The app has processed over USD 100 billion in cumulative transfers since launch.

Other fintech entrants attacked from different angles. Remitly focused on mobile first design for users whose primary device is a smartphone. Revolut bundled remittances into a broader financial app. In the UAE, companies like Denarii Cash targeted the specific corridor to South Asia, where volumes are enormous and fees were historically steep.

The clearest losers are the legacy operators. Western Union's stock price has declined significantly from its 2014 peak as transaction volumes shifted to digital channels. The company has attempted to build its own digital platform, but it carries the cost base of its physical network while competing with companies that never had one.

The second group of losers, less obvious, are the small exchange houses that operated in every neighbourhood. In the UAE, these shops served workers who were unbanked or underbanked, offering cash in and cash out services. Many have closed as customers shifted to apps. The workers who relied on them, often those least comfortable with technology, have been forced to adapt or find alternatives.

The winners are the senders and recipients. A worker sending USD 300 per month home now saves roughly USD 15 to 25 per transfer compared to the old model. Over a year, that is USD 180 to 300, which in many recipient countries is a meaningful sum, enough to cover a month of school fees or a medical bill.

The fintech disruption of remittances is one of the clearest examples of technology lowering costs for consumers by removing physical infrastructure and the margins it sustained. The total global cost of remittances has fallen from an average of roughly 10 percent in 2008 to under 6.5 percent in 2023, driven primarily by digital competition. However, the cheapest corridors are now digital to digital, meaning they serve people who have smartphones, bank accounts, and digital literacy. The most expensive corridors, often those serving the poorest senders, have barely moved. Fintech disrupted the industry but it has not yet finished the job, because the last mile of financial inclusion is not a technology problem. It is a trust problem.

Mark: 46/50

Excellent structural analysis of why legacy fees stayed high (branch networks as barriers to entry) and how fintech attacked the cost base by eliminating physical infrastructure. The distinction between visible fees and hidden exchange rate markup shows strong financial literacy.

To attain higher marks needed to explore the last mile trust problem with a specific example of a corridor where digital remittances have not penetrated.

Finance Prompt 2: Choose one country whose currency lost significant value in the last decade. Explain what caused the decline and how it affected ordinary people's purchasing power.

The Turkish Lira and the Cost of Unconventional Policy

In January 2014, one US dollar bought roughly 2.2 Turkish lira. By January 2024, the same dollar bought roughly 30 lira. The currency lost over 90 percent of its value against the dollar in a decade, making it one of the most dramatic currency collapses in a major middle income country in recent history. This essay explains the causes of the decline and traces its effect on the purchasing power of ordinary Turkish citizens.

Currency values are ultimately set by supply and demand in foreign exchange markets. Demand for a currency rises when foreign investors want to buy assets denominated in it (attracted by high real interest rates, strong growth, or political stability) and falls when they want to leave. The lira's collapse was driven by a combination of unconventional monetary policy, high inflation, and loss of investor confidence, each reinforcing the others.

The central mechanism was the relationship between inflation and interest rates. Orthodox monetary policy prescribes that when inflation rises, the central bank should raise interest rates. Higher rates make the currency more attractive to hold (because deposits and bonds pay more), which supports its value, and they cool domestic demand, which slows price increases.

Turkey's central bank did the opposite. Under political pressure, it cut interest rates repeatedly even as inflation accelerated, reaching a policy rate of 8.5 percent in early 2023 while inflation ran at over 50 percent. This produced deeply negative real interest rates: anyone holding lira was losing purchasing power faster than interest could compensate. The rational response for anyone with access to foreign currency was to sell lira and buy dollars or euros, which increased supply of lira in foreign exchange markets and pushed its value down further.

The falling currency then fed back into inflation. Turkey imports a large share of its energy and raw materials, all priced in dollars. A weaker lira made these imports more expensive in local terms, pushing up production costs, which were passed to consumers as higher prices. Higher prices prompted further selling of the lira, creating a self reinforcing cycle.

The most direct impact was on purchasing power. Turkey's official inflation rate peaked at over 85 percent year on year in October 2022, meaning prices roughly doubled in a single year. Independent estimates from groups like the Inflation Research Group (ENAG) suggested real inflation was even higher.

For a household earning a fixed salary in lira, the effect was devastating. A family spending 3,000 lira per month on groceries in 2020 faced a bill of roughly 8,000 to 10,000 lira for the same basket by 2023. Wages rose, but not at the same pace: the minimum wage was increased multiple times but consistently lagged the inflation rate, meaning real wages, the amount of goods and services a wage can actually buy, fell.

Housing costs were particularly affected. Rents in Istanbul rose by over 100 percent in some districts between 2021 and 2023. Property prices, often informally indexed to the dollar, rose even faster in lira terms. Young people attempting to buy their first home found themselves priced out by a combination of asset price inflation and mortgage rates that, when they finally rose, exceeded 40 percent per year.

Savings were destroyed. A Turkish citizen who held 100,000 lira in a savings account in 2014 had the equivalent of roughly USD 45,000. By 2024, the same 100,000 lira was worth roughly USD 3,300. Even with interest earned over the decade, the purchasing power loss was catastrophic.

The lira's collapse was not caused by war, sanctions, or natural disaster. It was caused by policy choices: keeping interest rates below inflation for an extended period, eroding confidence in the currency, and triggering a self reinforcing cycle of depreciation and price increases. The cost was borne overwhelmingly by ordinary citizens, particularly wage earners and savers, who had no ability to move their wealth into foreign currency the way corporations and wealthy individuals could. The Turkish case is a precise, measurable demonstration of how monetary policy affects purchasing power, and why currency stability is not an abstract concern but a direct determinant of living standards.

Mark: 45/50

Clear causal chain from negative real interest rates through currency depreciation to inflation, with the self reinforcing cycle well explained. The purchasing power examples (grocery bills, housing, savings erosion) make the impact tangible.

Could include more precise data on wage growth versus inflation to quantify the real wage decline. The essay focuses heavily on monetary policy and could briefly acknowledge other contributing factors such as current account deficits or political risk premium.

Finance Prompt 3: Pick one investment that most people consider safe (property, gold, government bonds, or another). Argue whether it is actually as safe as people believe, using evidence.

Is Property Really the Safest Investment?

In the UAE, as in much of the world, property is treated as the gold standard of safe investing. Parents tell their children to buy a flat. Financial advisors describe real estate as a hedge against inflation. The common belief is that property always goes up, and that even if it dips, you can always live in it. This essay argues that residential property is significantly riskier than most people believe, and that the perception of safety comes from psychological biases rather than financial evidence.

Property has real strengths as an investment. It is tangible: you can see it, touch it, and use it. It is scarce: land in desirable locations cannot be manufactured. It generates income if rented, and it provides utility (shelter) even when prices fall. Historically, property prices in many markets have risen over long periods, broadly tracking or exceeding inflation.

In the UAE specifically, property is often purchased without mortgage debt, which removes one of the largest risks (forced sale when loan payments cannot be met). Dubai's freehold property market, opened to foreign buyers in 2002, has attracted global capital, and the narrative of permanent demand growth driven by population increase and tourism is widely accepted. The evidence tells a more complicated story. Property is illiquid, concentrated, leveraged (for many buyers), and subject to drawdowns that can last for years.

Illiquidity means that selling a property takes weeks or months, compared to seconds for a stock or bond. In a falling market, the time to sell increases because buyers withdraw. During Dubai's 2009 to 2011 correction, property prices fell by roughly 50 percent from their 2008 peak, and transaction volumes collapsed. An investor who needed cash during that period could not exit at anything close to the pre-crisis price, and many could not exit at all.

Concentration is the risk of having too large a share of your wealth in one asset. A typical property purchase represents 50 to 90 percent of a household's net worth. No financial advisor would recommend putting 80 percent of a portfolio into a single stock, yet the equivalent concentration in a single property is considered normal. If that property's value falls, the household has no diversification to cushion the loss.

The "property always goes up" belief is contradicted by data from multiple markets. Japanese residential property prices peaked in 1991 and did not return to that level for over 30 years. Spanish property fell roughly 40 percent between 2008 and 2014. Irish property fell over 50 percent. Dubai's REIDIN residential price index showed a peak to trough decline of approximately 50 percent between 2008 and 2011, with full recovery not occurring until roughly 2023, fifteen years later.

Maintenance costs, service charges, insurance, and periods of vacancy (for rental properties) erode returns in ways that are rarely included in the "my property doubled" calculation. Studies

that compare property returns net of all costs to diversified stock market indices over 20 year periods typically find that equities outperform, with significantly better liquidity and lower transaction costs.

The perception of safety comes from at least three biases. First, survivorship bias: people remember the properties that rose in value and forget (or never hear about) those that did not. Second, money illusion: a property bought for AED 1 million and sold for AED 1.5 million twenty years later "made" AED 500,000, but after adjusting for inflation, maintenance, and the opportunity cost of capital, the real return may be negligible or negative. Third, the endowment effect: people value what they own more highly than what they do not, which makes property feel more solid than a number on a brokerage screen.

Property is a useful asset with real qualities: it provides shelter, it is tangible, and it has historically kept pace with inflation over very long periods. But it is not safe in the way most people use the word. It is illiquid, concentrated, subject to severe and prolonged drawdowns, and burdened by costs that are consistently underestimated. The perception of safety is driven more by psychology than by evidence. A genuinely safe portfolio would hold property as one component alongside diversified, liquid assets, not as the foundation.

Mark: 47/50

Confident, well evidenced argument that directly challenges a widely held assumption. The three biases (survivorship, money illusion, endowment effect) are correctly applied and the Dubai drawdown data is specific and relevant. Strong use of comparison with liquid alternatives.

The essay could acknowledge that property with no mortgage debt (common in the UAE) genuinely removes one of the largest risks, which would make the overall argument more nuanced. The maintenance cost point could use a worked example.

Engineering Prompt 1: Choose one city or country facing a serious water shortage. Explain the engineering solutions available and argue which one best balances cost, scale, and environmental impact.

Cape Town and the Engineering of Not Running Dry

In early 2018, Cape Town, South Africa, came within weeks of becoming the first major city in the modern era to run out of municipal water. The city announced "Day Zero," a date on which taps would be turned off and residents would queue at military guarded distribution points. Day Zero was ultimately averted through severe rationing, but the crisis exposed how fragile an urban water supply can be when demand outgrows infrastructure. This essay examines the engineering solutions available to water scarce cities and argues that managed aquifer recharge, combined with aggressive demand reduction, offers the best balance of cost, scale, and environmental impact.

Cape Town depends heavily on surface water stored in six reservoirs fed by winter rainfall. Between 2015 and 2017, the Western Cape experienced its worst drought in over a century: rainfall dropped to roughly 50 percent of the long term average for three consecutive years. Reservoir levels fell below 20 percent of capacity. The population of roughly 4 million people was placed under Level 6B water restrictions, limiting each person to 50 litres per day, roughly the amount used in a three minute shower and a few kettle fills.

The underlying vulnerability is concentration. Over 95 percent of the city's supply came from rainfall dependent surface reservoirs. When the rain failed, there was no backup. Four engineering approaches are available to cities facing water scarcity. Each has different cost, scalability, and environmental characteristics.

Desalination reverses osmosis to remove salt from seawater, producing drinking quality water regardless of rainfall. It is reliable and scalable: plants serving millions of people operate in Israel, Saudi Arabia, Australia, and the UAE. The drawbacks are energy consumption (roughly 3 to 4 kWh per cubic metre for reverse osmosis) and brine disposal. The concentrated salt stream returned to the sea can damage marine ecosystems if not properly dispersed. Capital costs are high, roughly USD 1 to 2 per cubic metre of capacity for a large plant, and operating costs depend heavily on energy prices.

Water recycling treats wastewater to a standard suitable for reuse. Direct potable reuse, where treated wastewater is returned to the drinking supply, is technically proven (Windhoek, Namibia, has practiced it since 1968) but faces public resistance. Indirect potable reuse, where treated water is injected into an aquifer and later extracted, is more widely accepted. Costs are roughly USD 0.5 to 1 per cubic metre, lower than desalination, and energy consumption is roughly half. Rainwater harvesting at the building level is low tech and low cost but cannot provide the volumes a city of millions requires. It is useful as a supplement, not a primary supply.

Managed aquifer recharge (MAR) involves capturing surplus water during wet periods, treating it to an appropriate standard, and injecting or infiltrating it into underground aquifers for storage and later recovery. The aquifer acts as a natural reservoir, with several advantages over surface

storage: minimal evaporation losses (surface reservoirs in hot climates can lose 10 to 30 percent of their volume to evaporation annually), protection from contamination, and enormous capacity without the land use of a dam.

For a city like Cape Town, whose problem is not a permanent absence of water but an unreliable pattern of rainfall, the best solution is one that stores surplus when it arrives and recovers it during drought. This makes MAR the strongest candidate.

The Cape Flats Aquifer, which sits beneath much of the city, has an estimated storage capacity of several billion cubic metres, far exceeding the total surface reservoir capacity of roughly 900 million cubic metres. Pilot projects have already demonstrated that treated stormwater and recycled wastewater can be injected into the aquifer and recovered at drinking water quality. The cost of MAR is favourable. Injection and recovery infrastructure costs roughly USD 0.1 to 0.5 per cubic metre, significantly below desalination. Energy consumption is low because the process relies on gravity and natural filtration rather than pressure driven membranes. Environmental impact is minimal if injection water is properly treated, because the aquifer provides additional natural filtration.

The limitation is that MAR requires a suitable aquifer, which not every city has. For cities without aquifers, desalination remains the reliable fallback despite its higher cost and energy use. Cape Town's near miss demonstrated that even water rich cities become water scarce when supply infrastructure is concentrated and inflexible. The engineering solution that best balances cost, scale, and environmental impact for cities with suitable geology is managed aquifer recharge: it is cheaper than desalination, more scalable than household harvesting, more publicly acceptable than direct potable reuse, and it turns the existing water cycle's variability from a vulnerability into a manageable problem. The key lesson is that the cheapest litre of water is often the one you already had and stored properly.

Mark: 46/50

Well structured comparison of four engineering solutions with clear cost, energy, and scalability data for each. The argument for managed aquifer recharge is convincing and properly qualified by the geological requirement. The Cape Flats Aquifer evidence is specific.

The essay could explore the political and institutional barriers to MAR adoption, which are often more significant than the engineering ones. The dismissal of rainwater harvesting is slightly too quick given its role in demand reduction.

Engineering Prompt 2: Pick one technology that exists today but is too expensive for widespread use (space travel, hydrogen fuel, lab grown meat, or another). Explain what engineering breakthroughs would need to happen to bring the cost down.

Why Green Hydrogen Is Still Too Expensive

Green hydrogen, produced by splitting water using electricity from renewable sources, is widely described as the fuel of the clean energy future. It produces no carbon emissions when burned or used in a fuel cell. It can store energy for months, power heavy trucks and ships that batteries cannot handle, and replace fossil fuels in industrial processes like steelmaking. Yet in 2024, green hydrogen costs roughly USD 4 to 8 per kilogram to produce, while grey hydrogen (made from natural gas, with carbon emissions) costs roughly USD 1 to 2. Until that gap closes, green hydrogen will remain a demonstration technology rather than a deployed one. This essay identifies the three engineering barriers that set the cost floor and explains what breakthroughs would bring it down.

An electrolyser splits water into hydrogen and oxygen using electricity. The dominant commercial technology is proton exchange membrane (PEM) electrolysis, which uses a solid polymer membrane coated with catalysts, typically platinum and iridium. These metals are extremely rare and expensive: iridium alone costs over USD 5,000 per ounce and global annual production is roughly 7 to 8 tonnes.

The breakthrough needed is a catalyst that performs comparably to iridium and platinum but uses abundant, cheap materials. Research into catalysts based on nickel, iron, cobalt, and manganese is advancing: some laboratory catalysts have achieved 80 to 90 percent of the performance of iridium at a fraction of the cost. The gap between laboratory demonstration and industrial scale deployment is the key engineering challenge: the catalyst must maintain performance for tens of thousands of hours, survive cycling on and off (as renewable electricity fluctuates), and be manufacturable at scale.

A second path is improving the membrane itself. Current PEM membranes degrade over time, requiring replacement every 5 to 10 years. Extending membrane life to 15 to 20 years would spread the capital cost over more hydrogen produced, reducing the cost per kilogram. Electrolysis is energy intensive: producing one kilogram of hydrogen requires roughly 50 to 55 kWh of electricity. At a solar or wind electricity cost of USD 0.03 per kWh (competitive in the best locations today), the electricity alone costs USD 1.50 to 1.65 per kilogram, already close to the total cost of grey hydrogen.

The breakthrough needed is further reduction in renewable electricity costs, which is primarily an engineering problem of manufacturing efficiency. Solar panel costs have fallen roughly 99 percent since 1976, and further gains are expected from perovskite tandem cells (which layer two light absorbing materials to capture more of the solar spectrum) and from thinner silicon wafers that use less material. If solar electricity costs fall to USD 0.01 to 0.02 per kWh, which some projections consider achievable by 2035, the electricity cost of hydrogen drops to under USD 1 per kilogram.

An electrolyser running at full capacity continuously produces the cheapest hydrogen per unit of capital invested. But solar and wind power are intermittent: a solar farm in the UAE operates at roughly 20 to 25 percent capacity factor (producing power roughly 5 to 6 hours per day on average). An electrolyser paired directly with solar therefore sits idle for 75 percent of the time, and the capital cost of the idle machine is spread over less output.

The engineering solution is either oversizing the renewable supply (so the electrolyser runs more hours, but at higher capital cost for the panels), adding battery storage (which adds its own cost), or connecting the electrolyser to the grid (which may mean using non renewable electricity at times). The optimal configuration is a systems engineering problem that depends on local solar resources, grid carbon intensity, and hydrogen storage costs. It does not have a single answer, but solving it for specific locations is tractable with today's tools.

Green hydrogen is too expensive for widespread use not because of a single barrier but because of the compounding of three: expensive electrolysers, energy intensive production, and intermittent renewable supply. Each barrier has a credible engineering path to resolution: abundant catalysts for the electrolyser, continued solar cost declines from improved cell architectures, and system level optimisation of electrolyser utilisation. If all three are achieved, production costs in favourable locations could fall below USD 2 per kilogram by the mid 2030s, making green hydrogen competitive with grey. The transition will not happen because hydrogen is clean. It will happen when it is cheap, and that is an engineering problem.

Mark: 45/50

Clear identification of the three compounding barriers with credible engineering pathways for each. The catalyst discussion shows genuine understanding of the gap between laboratory performance and industrial durability. Cost projections are realistic.

The utilisation barrier section could benefit from a worked example showing how capacity factor affects cost per kilogram. The essay could also briefly compare green hydrogen's trajectory to solar PV's historical cost decline to give the reader a sense of timescale.

Engineering Prompt 3: Choose one structure or machine built more than 100 years ago that still functions today. Explain what made it last and what modern engineers can learn from its design.

The Forth Bridge: 134 Years and Counting

The Forth Bridge, a cantilever railway bridge near Edinburgh, Scotland, was completed in 1890 and carries passenger and freight trains to this day. It spans 2,529 metres across the Firth of Forth, was the longest bridge in the world at the time of its construction, and used roughly 54,000 tonnes of steel and 6.5 million rivets. It remains in full operational service 134 years later, a fact that is not an accident but the direct result of specific engineering decisions that modern designers can still learn from.

The Forth Bridge was built from open hearth steel, a relatively new material at the time. The engineers, John Fowler and Benjamin Baker, chose steel over the wrought iron that had been standard for bridges because steel offered higher tensile strength, roughly 30 to 40 percent more than wrought iron, allowing longer spans with less material.

Crucially, they then overdesigned the structure by a significant margin. The Forth Bridge was designed to carry loads far exceeding any traffic anticipated at the time. This was not conservative timidity: it was a direct response to the Tay Bridge disaster of 1879, in which a poorly designed railway bridge collapsed in high winds, killing 75 people. The Tay failure had been caused partly by inadequate allowance for wind loading. Fowler and Baker responded by designing the Forth Bridge to withstand wind pressures of 56 pounds per square foot, roughly four times the pressure that had destroyed the Tay Bridge.

This overdesign created what engineers call a margin of safety: the gap between the maximum load the structure was designed for and the maximum load it would ever actually experience. The Forth Bridge's margin has proved large enough to absorb not only the original traffic but the heavier and faster trains of the modern railway network, even though these were inconceivable in 1890.

The second factor in the bridge's longevity is its design for maintenance. The structure is built entirely from riveted connections rather than welded joints. Riveted connections can be inspected individually, and a deteriorating rivet can be replaced without disturbing the surrounding structure. A welded joint, by contrast, is a continuous material bond that must be inspected using nondestructive techniques (ultrasound, radiography) and repaired by grinding out and rewelding.

The bridge's geometry also facilitates access. The tubular members that form the main structural elements are large enough for maintenance workers to climb inside and inspect from within. Every surface is accessible for painting and corrosion treatment. The famous claim that "painting the Forth Bridge" is an endless task reflects not poor design but deliberate design for continuous maintenance: the bridge was built assuming it would need painting forever, and the structure was shaped to make that possible.

The painting regime itself is instructive. The bridge was repainted continuously from 1890 to

2011, when a modern three coat epoxy system was applied that is expected to last 20 to 25 years, finally breaking the cycle. The original paint system required permanent repainting because the lead based paints available in 1890 had limited durability, but the critical point is that the designers assumed their protective coating would fail and designed the structure to tolerate repeated reapplication. They designed for the maintenance they knew would be needed, not the maintenance they hoped to avoid.

Three lessons from the Forth Bridge are directly applicable to modern design. The first is that designing for inspection and repair extends life more reliably than designing for perfection. Modern structures increasingly use welded steel and bonded composites that are stronger and lighter but harder to inspect and repair. The Morandi Bridge in Genoa, which collapsed in 2018, used stay cables encased in concrete that could not be visually inspected, a direct contrast to the Forth Bridge's philosophy of making every component accessible.

The second is that margin of safety absorbs the future. The Forth Bridge carries trains that are heavier and faster than anything its designers imagined, and it does so safely because the original margin was generous. Modern design culture, driven by material cost optimisation and computational precision, tends toward smaller margins. This is economically rational in the short term but fragile in the long term: a structure designed to exactly meet today's loads has no capacity for tomorrow's.

The third is that maintenance is a design decision, not an afterthought. The Forth Bridge was shaped so that every surface could be reached and every connection could be replaced. Modern buildings and infrastructure often treat maintenance access as a secondary concern, squeezed by cost pressure and aesthetic preference. The result is structures that look impressive when new but become difficult or impossible to maintain as they age.

The Forth Bridge has lasted 134 years because its engineers designed for uncertainty, maintainability, and generous margins rather than optimising for minimum cost. In an era when computational tools allow structures to be designed to exactly meet their specifications, the Forth Bridge is a reminder that the most durable engineering is the kind that assumes things will go wrong and leaves room. Its rivets can be replaced, its surfaces can be reached, and its capacity exceeds its load. These are not signs of waste. They are the reason it is still standing.

Mark: 46/50

Excellent use of a specific historical case to extract general engineering principles. The three lessons (inspectability, margin of safety, maintenance as a design decision) are clearly argued and the Morandi contrast is well deployed. The painting regime detail is insightful.

The essay could include more precise numbers on the original design loads versus current train weights to quantify the margin of safety claim. The concluding paragraph, while strong, slightly overlaps with points already made.